

Named Entity Recognition for Brazilian Portuguese Product Titles

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Abstract. Improving the interaction between consumers and marketplaces, focusing on reaching higher conversion rates is one of the main goals of e-commerce companies. Offering better results for user queries is mandatory to improve user experience and convert it into purchases. This paper investigates how named entity recognition can extract relevant attributes from product titles to derive better filters for user queries. We conducted several experiments based on MITIE and BERT applied to smartphones/cellphones product titles from the largest Brazilian retail e-commerce. Both of our strategies achieve outstanding results with a general F1 score of around 95%. We concluded that using a classical machine learning pipeline is still more useful than relying on large pre-trained language models, considering the model's throughput and efficiency. Future work may focus on evaluating the scalability and reusability capacity of both approaches.

Keywords: product titles · named entity recognition · MITIE · BERT · Portuguese.

1 Introduction

Improving the interaction between consumers and marketplaces, focusing on reaching higher conversion rates is one of the most significant challenges faced by e-commerce companies. In other words, the primary goal in this scenario is to guide the users to purchase the items they are searching in the platform. Therefore, due to the increasing specificity of search, the shopping experience becomes a bottleneck in this scenario, even for big e-commerce companies.

E-commerce shopping becomes a more daily activity than ever. In the pandemic scenario, Brazilian e-commerces saw an increase of 47% of purchases,

according to Ebit|Nelson 2020 report³. Besides, the product assortment, particularly for marketplaces, is getting more complicated since e-commerce is replacing traditional retail settings, like groceries and health care supply. A traditional marketplace strategy to improve product assortment is to introduce more sellers into the marketplace platform. On the one hand, it makes the consumers' lives easier, since they can acquire their needs in only one platform. On the other hand, it makes the product assortment to be increasingly non-uniform. Product titles broadly vary, either by their sellers or their nature/category.

Consider the following examples from our data to illustrate these problems:

1. *Smartphone Samsung Galaxy A01, 32GB, 2GB RAM, Tela Infinita de 5.7", Câmera Dupla Traseira 13MP (Principal) + 2MP (Profundidade), Frontal de 5MP, Bateria de 3000mAh, Dual Chip, Android - Azul*⁴
2. *Smartphone Moto G9 Play 64GB Dual Chip Android 10 Tela 6.5" Qualcomm Snapdragon 4G Câmera 48MP+2MP+ 2MP - Verde Turquesa*⁵
3. *iPhone SE 128gb Preto iOS 4G Wi-Fi Tela 4.7" Câmera 12MP + 7MP - Apple*⁶

The first title offers the following information: WIT⁷ (Smartphone); brand (Samsung); model (Galaxy A01); internal memory (32GB); display size (5.7"); quality of the cameras (13MP and 5MP); SIM card capacity (Dual chips) and color (Azul – "Blue" in Brazilian Portuguese). In the second example, the brand name is not explicit, but we can find the operational system (Android 10) and processor (Qualcomm Snapdragon) specifications. The last example shows: brand (Apple); model (SE); internal memory (128gb); color (preto – "black" in Portuguese); display size (4,7") and quality of the cameras (12MP and 7MP).

From these examples, we can see that the titles from a quite common category (Smartphone) do not follow a standard structure and are not even offering the same information. As a consequence, it is very difficult to specify standard search filters that could fit any title or query in this category.

In this scenario, increasing the products' retrieval becomes a difficult challenge. A syntax-based search engine is not enough to make all the assortment accessible to the customer. Commonly, e-commerces web pages offer facets to filter customer queries. It becomes a critical task for offering useful filters, having a uniform, vast, and high-quality product attribute extraction in place [11].

Hence, motivated by improving the human interaction with our e-commerce platform, we investigated how **named entity recognition** could be applied to

³ <https://company.ebit.com.br/webshoppers/webshoppersfree>

⁴ In English: Samsung Galaxy A01 Smartphone, 32GB, 2GB RAM, 5.7" Infinite Screen, 13MP Dual Rear Camera (Main) + 2MP (Depth), 5MP Front, 3000mAh Battery, Dual Chip, Android - Blue

⁵ In English: Moto G9 Play 64GB Smartphone Dual Chip Android 10 Screen 6.5" Qualcomm Snapdragon 4G Camera 48MP+2MP+2MP - Turquoise Green

⁶ In English: iPhone SE 128gb Black iOS 4G Wi-Fi Screen 4.7" 12MP + 7MP Camera - Apple

⁷ WIT stands for "What Is This" and is used to define the product being sold.

identify attributes in product titles of the Smartphone category⁸. We investigated two approaches: MITIE⁹ and BERT [6].

From our experiments we found that models trained with MITIE and BERT had very similar results, achieving around 95% in F1 scores. MITIE was chosen as the best one for our scenario due to other advantages such as its prediction latency and required infra-structure.

Thus, the main contributions of this work are:

- The description of an end-to-end approach of corpus annotation, model building and evaluation for attribute-value pairs extraction from product titles;
- The careful comparison of two state-of-the-art models for named entity recognition – MITIE and BERT – in a not well explored domain (e-commerce);
- The addressing of a poorly explored scenario of e-commerce language for Brazilian Portuguese.

This paper is organized as follows. In section 2 we describe the task of named entity recognition and the approaches proposed to perform it. In section 3 we describe the set of attribute products that we found relevant for our problem. The experiments and results are presented in section 4 and section 5 finishes the paper with some final remarks.

2 Background and related work

Named entity recognition (NER) is a subtask of natural language processing (NLP) or information extraction (IE) that tries to identify and classify “named entities” present in unstructured text. Named entities are well-known predefined categories such as people’s names (e.g., Bill), organizations (e.g., United Nations), locations (e.g., Brazil), time expressions (e.g., yesterday), quantities (e.g., liter), monetary values (e.g., dollar), among others. In the scope of e-commerce, our named entities will be renamed to **attributes**, and the attributes that are of our interest will be specified in section 3.

As noted by More [11], extracting attributes from product titles in e-commerce is a difficult task since titles, in this context, lack syntactic structure. For this purpose, the author proposes a NER-based method and demonstrates results on the particular attribute “brand”. More’s proposal consists of combining Structured Perceptron and Conditional Random Fields to predict BIO (B = begin, I = in, O = out) tags in product titles in English. The author used an automatically labeled (by regex) dataset to train the models and increased its volume and quality by a post-processing step, which includes augmenting a normalization dictionary and manual feedback from domain experts. The author reports a F1-score of 92% for the attribute “brand”.

⁸ The Smartphone category encompasses cell phones, smartphones and iPhones sold by the largest Brazilian e-commerce marketplace, B2W Digital.

⁹ <https://github.com/mit-nlp/MITIE>

Xu *et al.* [18] proposed an approach to support value extraction, scaling up to thousands of attributes with no loss of performance. The approach proposed by those authors uses: (i) a global set of BIO tags, (ii) a model of semantic representation for attributes and titles, and (iii) an attention mechanism to capture the semantic relations. The proposed model processes up to 8,906 attributes with an overall F1 score of 79.12%.

Cheng *et al.* [4] explored NER in the e-commerce context, proposing an end-to-end procedure to query annotation and extraction for English language, using BIO as a sequence tagging format and establishing two important entities to investigate: “brand” and “product type”. The annotation process was carried out with 16k queries. The proposed model reached a F1 score of 99.49% for the validation data and 93.30% for the test data.

Real *et al.* [14] presented the use of NER in the process of title generation in the fashion e-commerce context. The experiments showed that using a context-specific human-annotated data set for training, even though small (only 358 examples), achieved an overall F1 score of 71%. The solution proposed uses MITIE.

Recently, one of the greatest revolutions in NLP area was the development of the contextualized language models such as the Bidirectional Encoder Representations from Transformers (BERT) [6]. This language model encodes the text to a vector space where words with similar meanings are close to each other through a series of bidirectional self-attention heads. Unlike other well-known word embedding models, such as GloVe [13] and Word2Vec [10], BERT is capable of embedding words considering their specific contexts. BERT has been used successfully in several NLP tasks as an off-the-shelf solution. To achieve these results, BERT was trained in an unsupervised procedure, using *corpora* that sum up billions of words.

Souza *et al.* [15] trained a BERT-CRF to the NER task on the Portuguese language, and explored feature-based and fine-tuning training strategies for the BERT model. Their fine-tuning approach obtained new state-of-the-art results on the HAREM I dataset [1], improving the F1-score by 1 point on the selective scenario (5 Named Entities classes) and by 4 points on the total scenario (10 classes of Named Entities).

Besides BERT, we also investigate how the well-known MIT Information Extraction tool (MITIE) perform for the extraction of attributes from product titles. Geyer *et al.* [7] used MITIE to recognize named entities on a *corpus* consisting of 5,991 tweets in English, which have been annotated on seven different categories: person, organization, location, event, product, character, and thing. They varied the number of training documents and tested on the remainder of the documents, utilizing 5-fold cross-validation and only in-domain data. Their results showed that training with in-domain data began to show diminishing returns with 500 training documents across all of the entity types other than character and thing.

To the best of our knowledge this is the first work to investigate, in depth, the attribute-value extraction for the e-commerce domain in Brazilian Portuguese.

3 Attribute set

To define which attributes are relevant in Smartphone category, after an initial analysis of Smartphone titles, **10 attributes** were selected for being investigated in this work:

1. **WIT**: WIT stands for “What Is This” and is attached to the expression that better represents the product. In this work, the possible values for this attribute are: “smartphone”, “celular” (cell phone), “iphone” and other spelling variations. For example¹⁰:

Smartphone[wit] Samsung Galaxy S6 Desbloqueado Preto

2. **Brand**: The brand is one of the most relevant attributes in a customer query. Common values for this attribute are: “Motorola”, “Apple”, “Samsung”, “Xiaomi”, “Blu”, among others. For example:

Smartphone **Samsung**[brand] Galaxy J7 Prime Dual Chip Android 6.0 Tela 5.5” Octa-Core 1.6 GHz 32GB 4G Câmera

3. **Model**: The model is also an important attribute for queries in this category and should be extracted in its maximum extension. For example:

Sony **Xperia L2 H3321**[model] 32gb Lte 1sim Tela 5.5” Câm.13mp+8mp

4. **Internal memory**: The internal memory capacity is usually an important attribute for this category and must be extracted together with its unit measure in gigabytes (GB). Common values for this attribute are: 4GB, 8GB, 16GB, 32GB, 64GB, 128GB, 256GB and 512GB. For example:

iPhone 6S plus Apple **64GB**[internal_memory] Prata Seminovo

5. **Display size**: The display size should also be extracted with its unit measure in inches (”). The values for this attribute generally range from 4.7” to 6.1”. For example:

Smartphone Positivo Twist Pro, Dual Chip, Preto Tela **5.7”**[display_size], 3g+wifi, Android Oreo, 8mp, 32gb

6. **Processor**: Common values for this attribute are: “Dual-Core”, “Quad-Core”, “Hexa-Core”, “Octa-Core”, “Helio”, among others. For example:

Smartphone Alcatel Pixi4 5” Preto Com Dual Chip Memória 8gb Câmera 8mp 4g **Quad Core**[processor]

7. **Camera quality**: The quality of the smartphone camera is usually an attribute of great interest to customers. Subsequently, the resolution in megapixels (MP) must be extracted with its unit measure. The values for this attribute usually range from 5.0 MP to 18.0 MP. For example:

Smartphone Multilaser Ms50g Dual Chip Android 8.1 Tela 5.5 8gb 3g Câmera **8mp**[camera_quality] Bivolt

¹⁰ In this paper, the tags associated with the attributes are presented between brackets and the values associated to them, in bold.

8. **Color:** When the color of the product appears in the title, it must be extracted as a relevant attribute. For example:

Smartphone Nokia 6.1 dual Android 8.1 Tela 5.5 32GB 16MP - **Branco**[color]

9. **Operating system:** The operational system is an extremely relevant attribute for the smartphone category. The version of the operational system must also be extracted. Common values for this attribute are: “Android”, “IOS”, “Windows Phone”, “Blackberry OS”, among others. For example:

Smartphone Asus Zenfone GO Dual Chip **Android 5.1**[operational.system]
Tela 5” 16GB 4G Câmera 13MP - Dourado

10. **SIM card capacity:** When the SIM card capacity is present in the title, it must be extracted. The most common value for this attribute is “Dual Chip”. For example:

Smartphone Multilaser MS50S, **Dual Chip**[sim.card.capacity], Branco, Tela 5”, 3G+Wi-Fi, Android 6.0, 8MP, 8GB

These are the ten most relevant attributes for the Smartphone category that we found in our database. From this set, we built a small annotated sample and trained supervised named entity recognition approaches, aiming at automatically extracting attribute-value pairs from product titles for the Smartphone category as explained in the Section 4.

4 Experiments and results

In this section we describe the corpus used in our experiments and the annotation process (Section 4.1) carried out to feed the supervised named entity recognition (NER) approaches we investigated in this work. These approaches are detailed in Section 4.2 and their results presented in Section 4.3.

It is worth mentioning that we used the source code available in the official open repositories of each specific tool applied here and that the corpus is not freely available due to the americanas s.a.’s police we worked with.

4.1 Corpus pre-processing and annotation

To analyze the viability of NER to extract attributes from product titles of the Smartphone category, we defined a criterion to select a subset from the original data set of around 7 million products of general categories. We removed from the data set all the products with WIT not labeled as “smartphone”, “celular” (cell phone), or “iphone”. This selection resulted in a data set of 7,432 products.

From this set, we extracted only the product titles. The titles were, then, pre-processed by performing the following tasks: (i) Conversion of the text to lowercase; (ii) Removal of special characters such as: bars, parentheses, brackets and hyphens; (iii) Tokenization by white space; (iv) Stop words¹¹ removal; and (v) Removal of duplicate titles.

¹¹ Stop words list in Portuguese: de, a, o, que, e, do, da, em, os, no, na, por, as, dos, ao, das, à, ou, às, com.

Then, we removed the outliers¹² by excluding all titles that did not contain “smartphone”, “celular” or “iphone” and some possible misspelling variations (such as “smatphone”). In the end, 58 outliers¹³ were removed. Most of them were battery cells, biological cells or money bills probably wrongly categorized as “celular” due to the orthographic similarity between the word “celular” (cell phone) and the words “célula” (cell) or “cédula” (bill).

The remaining 7,374 product titles were then clusterized. To start the clustering task, the tokens “smartphone”, “celular” and related spelling variants were removed from the data set¹⁴. Then, the titles were vectorized using the TF-IDF vectorizer, using default parameter values. Using the Elbow Method, the number of clusters was set to be 16 and the Mini Batch K-Means clustering method was executed to group the product titles of the same brand, with a few deviations. The clustering was performed using scikit-learn [12].

From each cluster, 26 items were selected: 13 being the closest to the centroid and 13 being other ones randomly selected. As a result, a total of 416 product titles were selected to be annotated by two linguists. The annotation was performed using the Prodigy [3] tool. A third linguist was responsible for resolving the discrepancy cases. The percentages¹⁵ of agreement between the two annotators can be seen in Table 1.

The annotation of the title “Smartphone blu grand m dual sim 3g 5.0” 5mp 3.2mp prata” is illustrated in Figure 1 using the Prodigy [3] tool. As a result, the following attributes were marked: WIT (Smartphone), brand (blue grand m), SIM card capacity (dual sim), internal memory (3g), display size(5.0”), camera quality (5mp 3.2mp) and color (prata).

Fig. 1. Annotation example using the annotation tool Prodigy [3]



¹² Outliers refers to those product titles that do not belong to the Smartphone category.

¹³ Examples of removed outliers, in Portuguese: “célula vegetal ampliada aproximadamente 20 mil vezes”, “celula carga 10 kg sensor peso arduino” and “cédula foleada ouro 100 euros coleção notas moedas euro”.

¹⁴ We didn’t remove the token “iphone” and its spelling variations since the format of iPhone titles is very different from the other items in the data set and the experiments removing “iphone” led to results worse than those not removing it.

¹⁵ Since how to measure the inter-annotator agreement of named entities annotation is a debatable task [5], we followed [1] and considered the percentages of agreement as our main metric.

Table 1. Agreement between the two annotators for each attribute

	Equal	Different	Intersection	Total
WIT	386 (93.23%)	8 (1.94%)	20 (4.83%)	414
Brand	374 (92.41%)	18 (3.93%)	14 (3.66%)	406
Model	314 (76.96%)	25 (6.13%)	69 (16.91%)	408
Internal memory	225 (85.51%)	18 (6.89%)	20 (7.60%)	263
Display size	188 (76.11%)	12 (4.87%)	47 (19.02%)	247
Processor	28 (73.68%)	9 (23.68%)	1 (2.64%)	38
Camera quality	164 (88.17%)	9 (4.85%)	13 (6.98%)	186
Color	291 (91.79%)	16 (5.04%)	10 (3.17%)	317
Operating system	131 (92.25%)	7 (4.93%)	4 (2.82%)	142
SIM card capacity	207 (85.90%)	31 (12.86%)	3 (1.24%)	241

As Table 1 makes evident, WIT was the attribute with the highest agreement, being labeled equally in 386 of the 414 instances. On the other hand, Processor, Model and Display size were the attributes with the highest disagreement between annotators, with about 73% to 76% of the instances annotated equally.

4.2 Experimental Setup

In this section, we describe the tools for NER investigated in this work: BERT and MITIE. BERT is a recent approach which has achieved significant results in many NLP tasks [6]. MITIE, in turn, is an earlier well-known and widely used tool for NER and binary relation extraction. While the former is a neural network based approach, which relies on contextualized word vectors, the latter applies traditional machine learning algorithms and static word embeddings to identify named entities.

BERT (Bidirectional Encoder Representations from Transformers) [6] is a neural approach which outputs deep contextualized language representations derived from unlabeled text. Although training a big BERT model from scratch can be too computationally expensive, pre-trained BERT models can be fine-tuned to several NLP tasks without substantial task-specific architecture modifications.

Some freely available, pre-trained BERT models are: (i) the BERT-base Multilingual Cased¹⁶, which is a pre-trained model for 104 languages including Portuguese; and (ii) BERTimbau-Base¹⁷ [16], which was trained with the Brazilian Portuguese corpus brWaC [17], widely used in open-source projects.

Besides BERTimbau-Base, some pre-trained models are also available for NER [15], which use BERTimbau-Base as a start point and were optimized for NER using HAREM¹⁸ dataset. Two approaches were applied for NER by the BERTimbau authors: the feature-based and the fine-tuning.

¹⁶ <https://huggingface.co/bert-base-multilingual-cased/tree/main>

¹⁷ <https://github.com/neuralmind-ai/portuguese-bert>

¹⁸ <https://www.linguateca.pt/HAREM/>

In the feature-based approach, a pre-trained model is used as a starting point, but the weights for the BERT’s network layers did not change. A Long Short Term Memory (LSTM) layer, when possibly combined with a Conditional Random Fields (CRF) layer, is trained and used to classify the sentences.¹⁹ In the fine-tuning approach, a pre-trained model is also used as a starting point, but its weights are adjusted during the training to achieve better results for NER. Following the fine-tuning approach, it is possible to add a CRF layer at the end of the network to help with sentence classification.

In this work we fine-tuned two models for NER in our dataset: one fine-tuning only with BERTimbau-Base (BERTimbau) and another one starting from the BERTimbau-Base trained for NER with HAREM dataset and also adding a CRF layer (BERTimbau-HAREM).²⁰

To evaluate the robustness of these methods to random variations, two random seeds (`seed1` and `seed2`) were extracted from the data set for generating training (50%), validation (25%) and test (25%) partitions. In total, 4 models were trained using the BERT approach: BERTimbau in `seed1`, BERTimbau in `seed2`, BERTimbau-HAREM in `seed1` and BERTimbau-HAREM in `seed2`. For both pre-trained BERT models, the hyperparameters used were the default values.

MITIE (MIT Information Extraction) framework²¹ [9] was designed to be a simple, easy and intuitive tool with no parameters. The current version of MITIE includes tools for performing NER, binary relationship extraction, as well as tools for training custom models.

The MITIE NER model uses the WordRep benchmark for building word representations (word embeddings), word morphology and traditional machine learning algorithms such as Structural Support Vector Machines [8]. MITIE comes with some previously generated word embeddings for English, Spanish and German. In our experiments we generated new word embeddings, using WordRep, which is publicly available in the MITIE repository [2]. We used a dataset containing around 7 thousand titles from the Smartphone category. This word representation was called `celular`.

The same seeds (`seed1` and `seed2`) used for training BERT NER models were also used for training MITIE NER models with one difference: MITIE does not explicitly use a validation set. Thus, the seeds were divided only into training and test, following the proportion of 75% for training and 25% for testing. In this way, the training instances of MITIE correspond to the training and the validation instances used by BERT, and the test cases are the same for both models.

¹⁹ From previous experiments, we concluded that the feature-based models as well as the ones trained with the multilingual BERT did not reach results as good as the fine-tuned models trained with BERTimbau-Base, so these options are not described here.

²⁰ These two configurations were the ones which achieved the best results in previous experiments with product tiles from Fashion category.

²¹ <https://github.com/mit-nlp/MITIE>

In addition to the `celular` word representation created in this work, existing ones for English (`English`) and Spanish (`Spanish`) available with MITIE were also used. We also generated a huge word representation from 7.5 million titles for products from diverse categories (not only Smartphone). This representation was called `all`. Finally, we also generated two versions of the `celular` word representation removing the titles from `seed1` (`cel-seed1`) and `seed2` (`cel-seed2`). Ergo, ten different models were trained in total using these 6 word representations.

4.3 Results

In this section we present the results for BERT and MITIE separately, then a comparison of both approaches will be made in the next section by drawing some conclusions about their performances on solving the investigated problem.

BERT Results As described in section 4.2, two BERT models were fine-tuned for NER (attribute extraction) in product titles: the original BERTimbau (BERTimbau) and the BERTimbau trained for NER with the HAREM dataset and a CRF layer (BERTimbau-HAREM).

Table 2 shows that the BERT models present F1 scores of around 94-95%, proving to be very robust independently of the random seed, indicating the absence of overfitting.

Table 2. F1 scores for BERT models in test set

		BERTimbau	BERTimbau-HAREM
seed 1	Micro-AVG F1	0.9394	0.9325
	Macro-AVG F1	0.9399	0.9323
seed 2	Micro-AVG F1	0.9496	0.9387
	Macro-AVG F1	0.9495	0.9382

It is interesting to notice that, although by a little difference, the model fine-tuned from the BERTimbau-Base performed better than the one fine-tuned from a version trained for NER in HAREM dataset.

Table 3 provides the detailed results for the best scenario: the BERTimbau model in the test set of `seed2`. The “support” column indicates the amount of evaluated instances in each class. The classes with the worst F1 scores were *Model* and *Processor*. The former one, probably due to labelling errors for the boundaries of the entity and the latter one maybe due to the small amount of training instances (less than 40).

MITIE Results As described in section 4.2, ten MITIE models were trained using 6 different word representations: `English`, `Spanish`, `all`, `celular`, `cel-seed1` and `cel-seed2`. Table 4 shows the F1 scores for these models.

Table 3. Detailed evaluation of BERTimbau fine-tuned with **seed2** in the test set

	precision	recall	F1	support
WIT	1.0000	0.9905	0.9952	105
Brand	0.9898	0.9700	0.9798	100
Model	0.8224	0.8381	0.8302	105
Internal memory	0.9722	0.9859	0.9790	71
Display size	0.9571	0.9853	0.9710	68
Processor	1.0000	0.7500	0.8571	12
Camera quality	0.9394	0.9841	0.9612	63
Color	0.9778	0.9462	0.9617	93
Operating system	0.9744	0.9048	0.9383	42
SIM card capacity	0.9552	0.9697	0.9624	66
Micro-AVG	0.9515	0.9476	0.9496	725
Macro-AVG	0.9525	0.9476	0.9495	725

Table 4. F1 scores for the MITIE models in test set

		English	Spanish	all	celular	cel-seed1	cel-seed2
seed 1	Micro-AVG F1	0.9359	0.9324	0.9290	0.9325	0.9351	–
	Macro-AVG F1	0.9352	0.9300	0.9328	0.9335	0.9306	–
seed 2	Micro-AVG F1	0.9396	0.9390	0.9378	0.9453	–	0.9397
	Macro-AVG F1	0.9428	0.9448	0.9441	0.9492	–	0.9466

From F1 scores we can see, again, that the values obtained for **seed1** and **seed2** are very close – again around 94-95%. Regarding the wordreps (word embeddings), the **English** and **celular** were the ones with best results and removing the titles from **seed1** or **seed2** did not significantly impact the results. Thus, it is possible to conclude that having the titles used in training for the generation of the word representations did **not** significantly bias the model. Therefore, the model trained with **celular** for **seed2** was the one chosen for further analysis.

In order to do that, Table 5 presents the detailed results for the evaluation of the MITIE model trained with **celular** and **seed2** in the test set. It can be noticed that, as happened with the BERT models, the MITIE model also performed poorly in the *Model* class. Again, we believe that this is due to entity labelling errors.

Example To finish our results and discussion section, Table 6 shows the MITIE’s predictions output by the model trained with **celular** and **seed2** for the three product titles:

1. *smartphone samsung galaxy j6, 32gb, dual chip, android, tela 5.6 pol, octa core, 4g, 13mp, tv prat;*
2. *iphone 7 apple, ouro rosa, tela 4.7”, 4g+wifi, ios 11, 12mp, 32gb;*
3. *celular up 3 chip quad cam mp3 4 fm preto cinza p3274 multilaser*

Table 5. Detailed evaluation of the MITIE using `celular` and `seed2` in the test set

	precision	recall	F1	support
WIT	0.9905	0.9905	0.9905	105
Brand	0.9706	0.9900	0.9802	100
Model	0.8462	0.8381	0.8421	105
Internal memory	0.9571	0.9437	0.9504	71
Display size	0.9848	0.9559	0.9701	68
Processor	1.0000	0.9167	0.9565	12
Camera quality	0.9677	0.9836	0.9756	61
Color	0.9140	0.9140	0.9140	93
Operating system	1.0000	0.9048	0.9500	42
SIM card capacity	0.9420	0.9848	0.9636	66
Micro-AVG	0.9472	0.9433	0.9453	723
Macro-AVG	0.9573	0.9422	0.9492	723

Table 6. Examples of MITIE’s predictions using `celular` and `seed2`

	title 1	title 2	title 3
WIT	smartphone	iphone	celular
Brand	samsung	apple,	multilaser
Model	galaxy j6,	7	“up”, “4 fm”
Internal memory	32gb,	“4g+wifi”, “32gb”	-
Display size	5.6 pol,	4.7”,	4 fm
Processor	octa core,	-	quad
Camera quality	13mp,	12mp,	-
Color	prat	“ouro”, “rosa”	“preto”, “cinza”
Operating system	android,	-	-
SIM card capacity	dual chip,	-	3 chip

The only differences between BERT and MITIE’s outputs for these examples were: for title 1, BERT did not output a value for *Color* attribute; for title 2, in the *Internal Memory*, the only output of BERT was “32 gb” and the “ios 11” was tagged as *Operating system*. For title 3, “quad” token was tagged as *Brand*, the *Display size* attribute was left empty, and the *Model* attribute values were “up” and “p3274”.

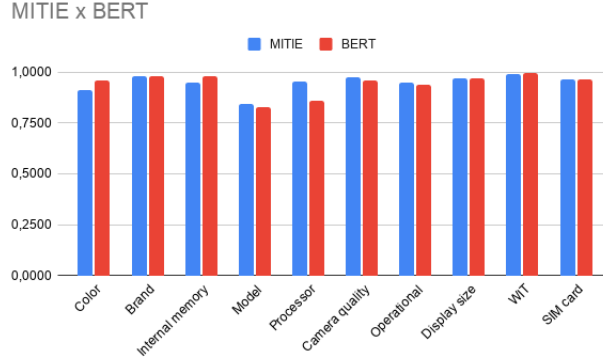
4.4 Comparison between the models

Figure 2 shows a graph of F1 scores for the best BERT (BERTimbau in `seed2`) and MITIE (trained with `celular` and `seed2`) models.

As we can see from Figure 2, the performances are very close with little highlights to: BERTimbau in *Color* class and MITIE `celular` in *Processor* class. In fact, it is worth highlighting the good performance of the MITIE model in the *Processor* class, which has only less than 40 training instances.

Regarding the **training time**, we can point out that, when using the CPU, both MITIE and BERT took a considerable amount of time, however **MITIE**

Fig. 2. Comparison of F1 scores for the best models (BERTimbau and MITIE trained with `celular`) trained with `seed2` in the test set



was still faster than BERT. BERT took around 7 hours against 5 hours from MITIE to train each model. However, the BERT training can be speed up using GPUs as accelerators: **with a median GPU, the fine-tuning of BERTimbau took only 28 minutes.**

Regarding the **prediction latency** (that is, the time for inferring new entities), the BERTimbau model took **97 minutes and 15 seconds** to infer all the entities in the dataset containing 7,374 titles running (only on CPU) in a machine with a Intel(R) Xeon(R) Silver 4210 CPU @ 2.20GHz processor and 40 cores. Using a GPU Gigabyte NVIDIA GeForce RTX 2060 Windforce OC (6GB), the time needed to finish all inferences decreased considerably: the model took only **11 minutes e 52 seconds** to complete the inference for the same set of 7,374 instances.

On the other hand, for the same dataset and the same CPU machine, the MITIE model trained with `celular` and `seed2` took approximately **3.64 seconds** to make all the inferences, showing an extremely better throughput when compared to BERT model. This mean 4 seconds from MITIE $\times$ 11 minutes from BERT. So, MITIE offers not only accuracy, but also efficiency as had already been pointed out as one of its great advantages by its developers.

Regarding the **size of the model** when stored on disk, BERT uses an average of 1.3 GB to store the model as a whole. For MITIE, the storage size of the model varies a lot and it is directly proportional to the size of the word representation used. For **English, Spanish** and all the model sizes are around 338 MB, for `celular`, `cel-seed1` and `cel-seed2`, the final size of the models is about 29 MB. In light of that, **the MITIE model wins again when the size of the model is considered.**

In general, both BERT and MITIE models achieved satisfactory results on the pursued task. The difference in F1 score between these methods is not significant. However, a few points need to be considered. First, MITIE performs better where BERT achieved the worst scores. Specifically in the processor and

model classes. In the classes where BERT performs slightly better than MITIE, the differences are minor, and the scores obtained by MITIE were considered adequate. Moreover, the computational resources required by MITIE are notably lower than those demanded by BERT. Considering these points and, particularly, the fact that the prediction latency is crucial for the application, we considered the **MITIE model trained on celular and seed2** as the most appropriate to generate the standardization dictionary that will give rise to search filters for the Smartphone category.

5 Final Remarks

This paper addressed the problem of attribute-value extraction from product titles of the Smartphone category in Brazilian Portuguese. By automating this process, the e-commerce companies are able to have a unified treatment to their product assortment, improving query matching, product recommendation and the use of facets to enhance product discoverability.

Several models were investigated in this paper based on BERT and MITIE approaches. After a careful analysis based on accuracy and efficiency, we concluded that, for our scenario, a solution based on MITIE is still more interesting than one based on pre-trained language models.

As future work, we want to investigate how scalable our approach is: would a MITIE-based model still get competitive accuracy when applied to different categories? Considering a sparsely annotated data scenario, which approach would be quick to deploy, keeping high-quality predictions? All in all, a question that remains is, across the 35MI products americanas s.a. sells now, will a classical machine learning approach still be competitive with deep learning approaches?

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